



Using Boolean Values

Practical Skills Module 2: Topic 5

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Topic Code	PM-02-PS05:
Topic	Declare a Boolean – true or false

Scope of Practical Skill

Given an applicable instruction and access to a learning platform, the learner must be able to:

- PA0401 Use double data types in Python

Applied Knowledge

- AK0401 Concept, definition and functions of Python data types

Internal Assessment Criteria

- IAC0401 Expected results with Python double are achieved

Scope of Practical Skill

Given an applicable instruction and access to a learning platform, the learner must be able to:

- PA0501 Declare a Boolean
- PA0502 Define true or false keywords
- PA0503 Compare two values
- PA0504 Evaluate two variables
- PA0505 Create functions that returns a Boolean Value

Applied Knowledge

- AK0501 Concept, definition and functions of Python data types

Internal Assessment Criteria

- IAC0501 Expected results with Python Boolean are achieved

Resources:

Knowledge	Information from Python Programmer Curriculum
National Curriculum Framework	900221-000-00-PM-02



Practical task

Follow the facilitator's instructions to complete the following activities:

Scenario: Library Membership System - Validating Borrower Eligibility

In this scenario, you are tasked with creating a Python program for a library membership system. The program uses Boolean values to determine whether a user is eligible to borrow a book, has overdue books, or meets specific criteria for borrowing. Here's how the steps align with the scenario:

Create a Python file called **borrower_eligibility.py** and complete the following steps:

Step 1: Declare a Boolean to represent membership status

- Create a variable to store whether the user is a **registered library member**.
- Assign either `True` or `False` to the variable based on the user's membership status.
 - *Example idea:* A member should have a value of `True`.

Step 2: Use `True` or `False` to indicate overdue books

- Create another Boolean variable that represents whether the user has **any overdue books**.
- Assign the appropriate `True` or `False` value to reflect that status.

Tip: If a user has overdue books, set the value to `True`; otherwise, `False`.

Step 3: Compare values to validate a membership ID

- Simulate a valid **membership ID** by assigning a reference value (e.g., `'ABC123'`).

- Create a variable that stores the user's entered ID (you can hardcode or use `input()`).
- Use a comparison operator to check if the user-entered ID matches the valid one.
 - *Expected output:* `Membership ID valid: True or False`

Step 4: Evaluate multiple Boolean conditions

- Check whether the user is allowed to borrow a book:
 - The user must **be a member** (True)
 - The user must **not have overdue books** (False)
- Use a logical operator (`and`, `or`, `not`) to combine the conditions and print the result.
 - *Example output:* `User eligible to borrow: True`

Step 5: Create a function that returns a Boolean

- Write a function named `can_borrow_book()` that accepts:
 - Membership status
 - Overdue book status
 - Membership ID check result
- The function should return `True` if **all conditions** for borrowing are met.
- Otherwise, return `False`.

Tip: Use an `if` statement or a return statement with logical conditions inside the function.

- Call the function and print the result.
 - *Expected output:* `Final eligibility status: True or False`



Take note

Important: Be sure to upload all files required for the task submission inside your task folder and then click "Request review" on your dashboard.
